

DistillToMCU

Cloud LLM

DeepSeek / Qwen
structured tool-call
decisions per event



COMIC distillation

online quantiles
conformal calibration
MDL consolidation



ESP32-S3

local rule matching
1.48 ms (p50)
no LLM at runtime

temp in [22.5, 28.0]
conf 0.92 · 2 B/rule

85.4% held-out autonomy

50.0% precision on UCI replay

teacher shift 1.7–3.7 pt